

AURORA + Decoders

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Lightweight Decoder

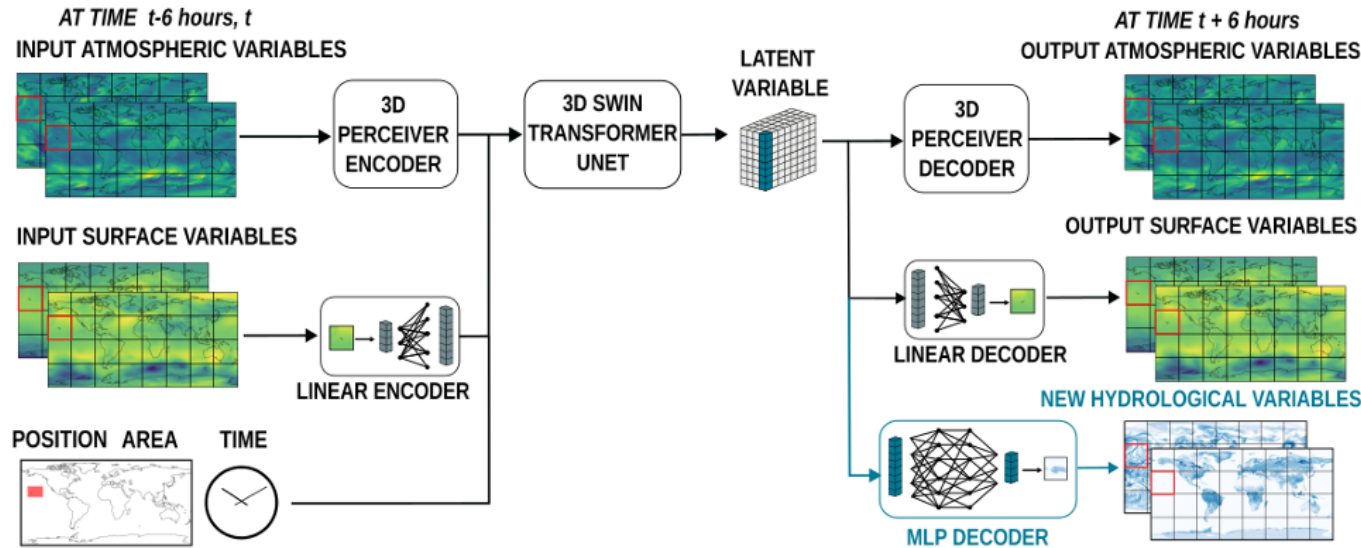


Figure 1: In black, the original Aurora model with its encoder-processor-decoder architecture. These blocks are frozen in our implementation. In blue, the additional Multi-Layer Perceptron (MLP) decoders that are trained to predict new physical variables.

- Lehmann (2025) came up with a MLP decoder that uses latent variables from the SWIN Transformer to predict new variables
- **Precipitation**, potential evaporation, runoff, and soil water content
- *Note that precipitation was trained on MSWEP precipitation data to avoid known biases of ERA5 precipitation data*

Decoder Performances

	Decoder	Aurora ⁺
FLOPS	4×10^{11}	31×10^{12}
samples/sec	0.34	0.16
allocated GPU memory	65GB	99GB

- Much more efficient in terms of computations

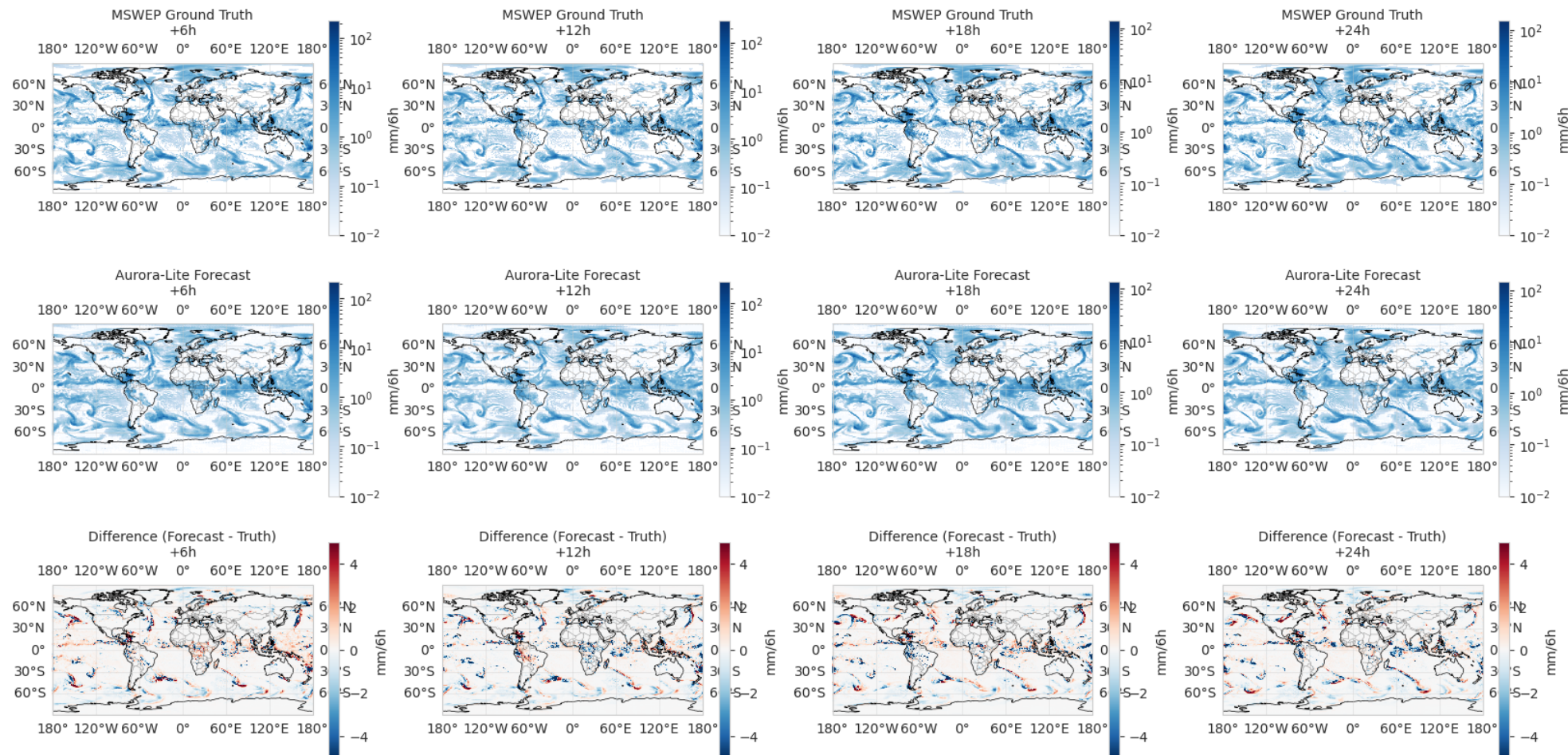
	Decoder	Aurora ⁺	IFS	GraphCast	FuXi
MAE (mm)	<u>0.32</u>	0.22	0.44	0.38	0.36
RMSE (mm)	<u>1.40</u>	1.03	1.60	1.50	1.42
SEEPS	0.35	0.23	0.34	0.33	<u>0.31</u>
W1	0.246	0.17	0.06	<u>0.15</u>	0.18
W1 log	0.046	<u>0.034</u>	0.029	0.047	0.042
bias (mm)	-0.14	<u>-0.096</u>	0.10	-0.02	-0.02
FSS 1mm	<u>0.92</u>	0.97	0.89	0.90	0.91
FSS 5mm	<u>0.82</u>	0.93	<u>0.82</u>	0.80	0.81
PCC	<u>0.71</u>	0.86	0.68	0.70	<u>0.71</u>

- Decoder and AURORA⁺ outperforms other models in most cases
- Despite AURORA⁺ generally performing better, decoder still succeeds to make acceptable forecasts

2020 6h lead time forecast metrics

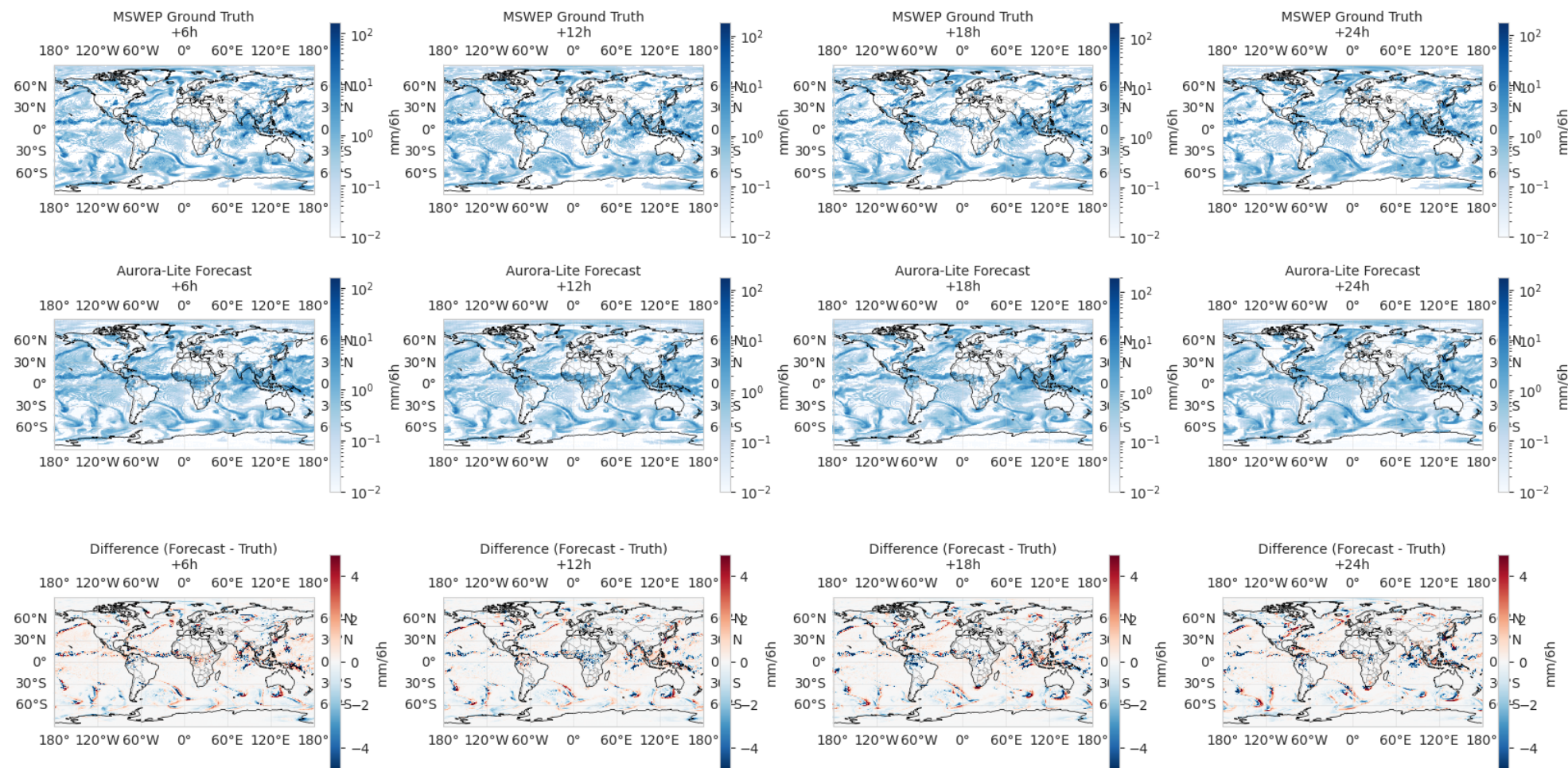
Trying the Decoder for Seen Data

Trained on: 17000 samples between
May 2002 and June 2014
Validated on: 1700 samples between
July 2014 and December 2015



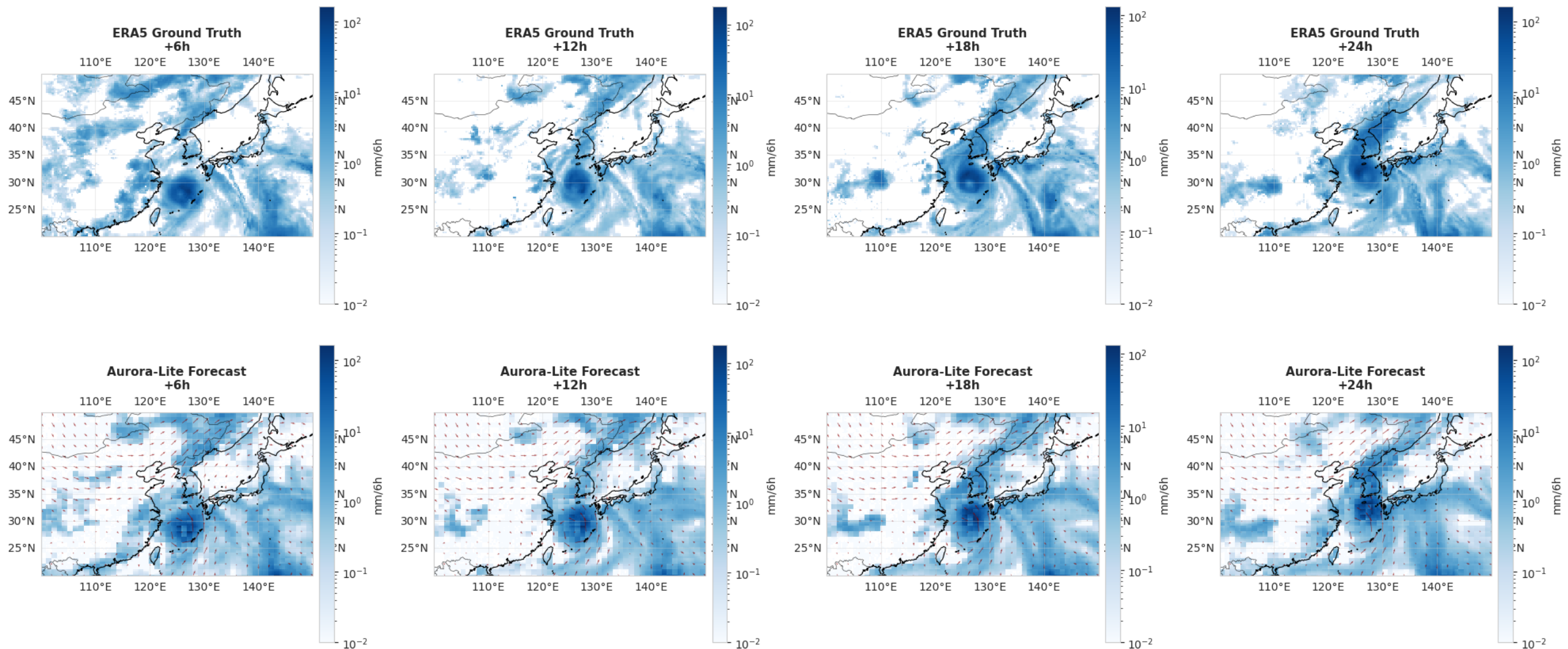
2012-10-24

Trying the Decoder for Unseen Data

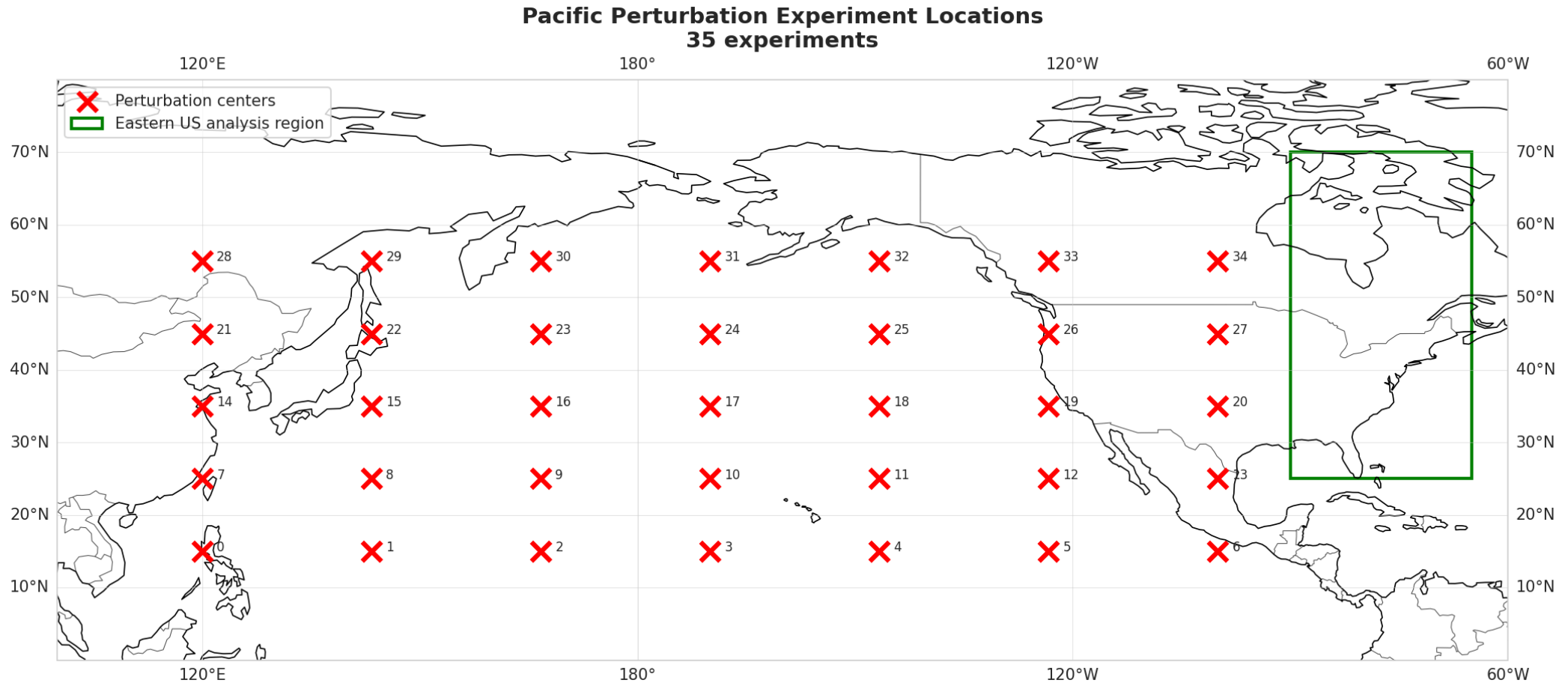


2020-09-01

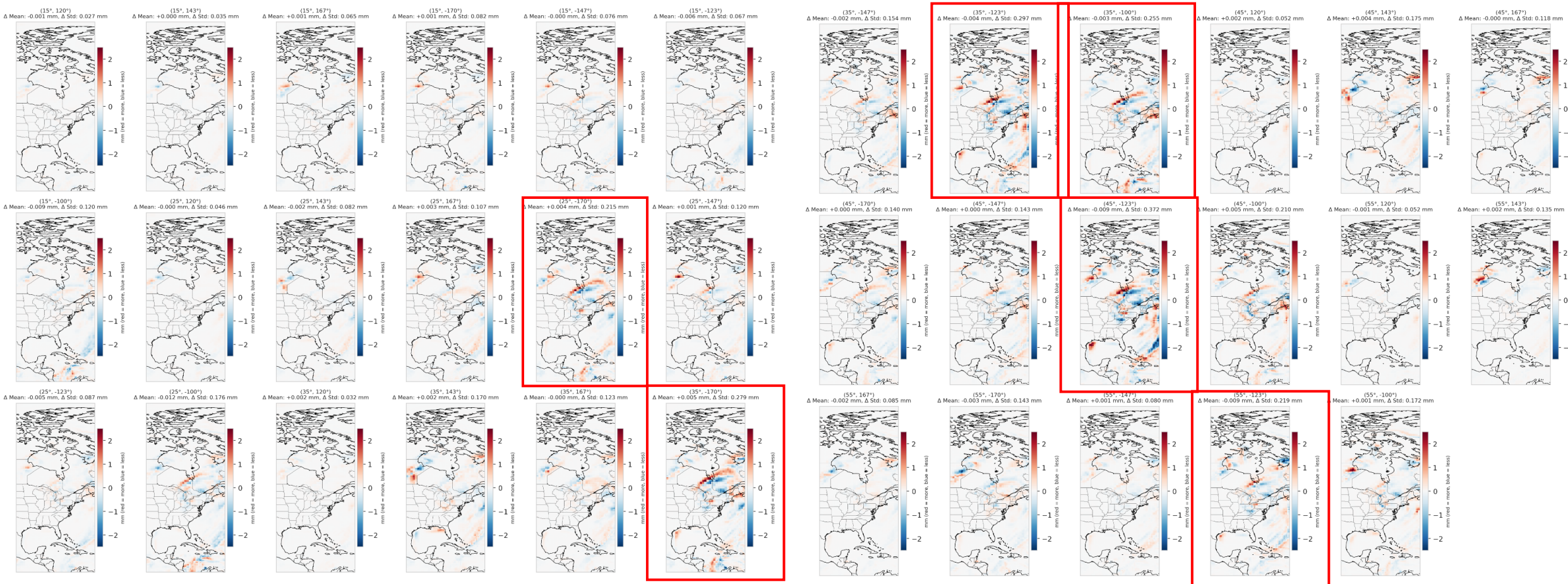
East Asia Precipitation with 500 hPa Wind Vectors: 2020-09-01 (UNSEEN DATA)



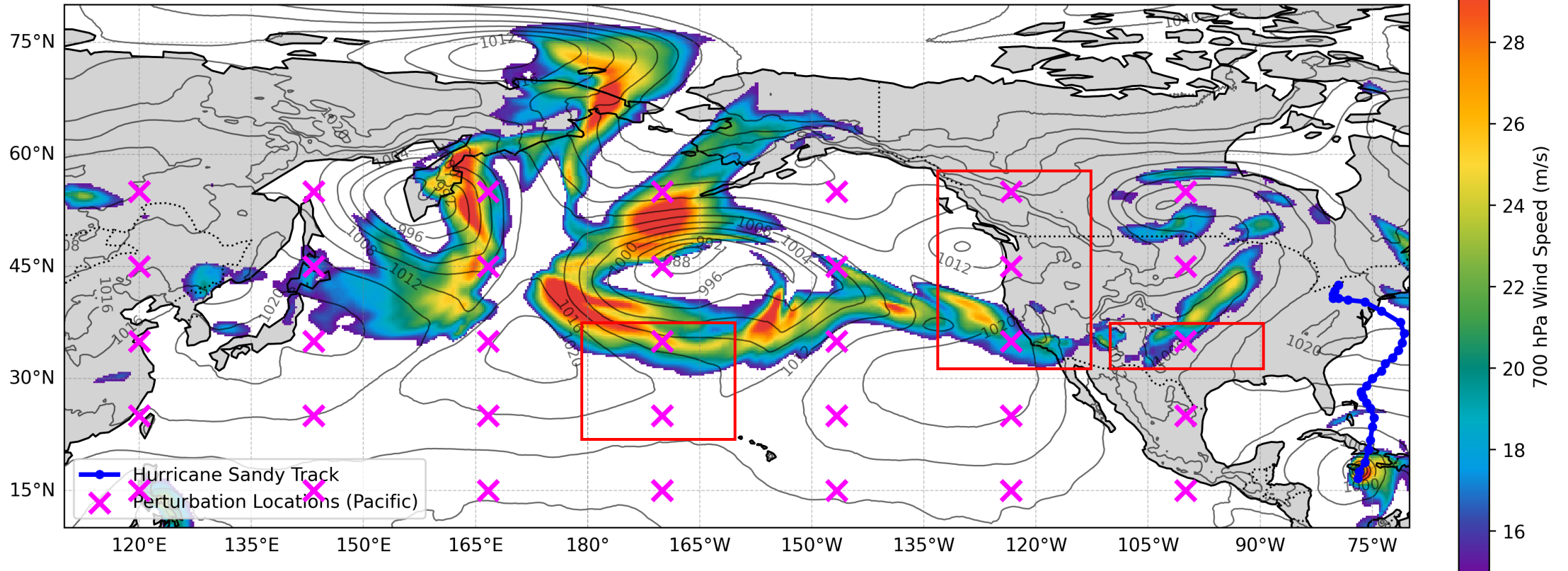
Decoder and Single Point Perturbation



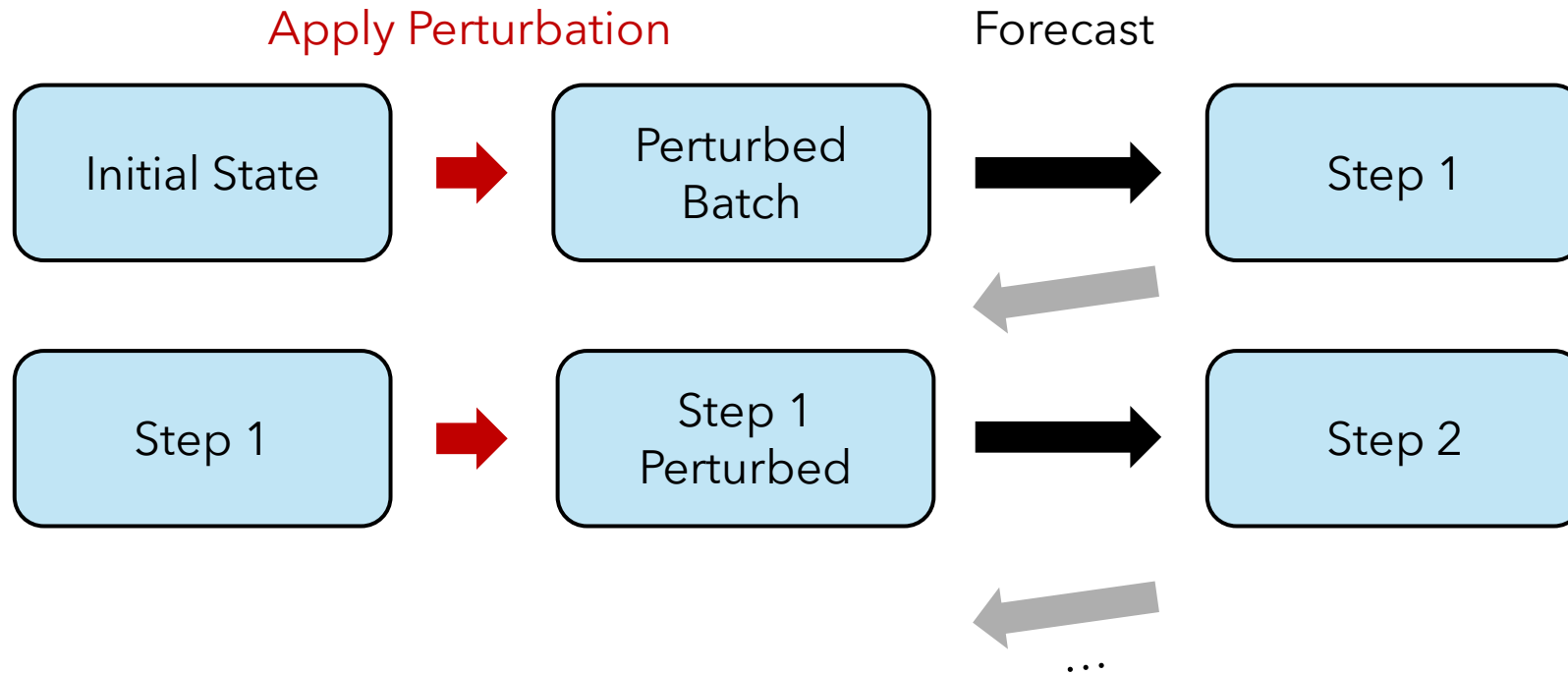
Eastern US Precipitation DIFFERENCE - Final Timestep
Perturbation – Control | 35 perturbations | Color scale: [-2.5, 2.5] mm

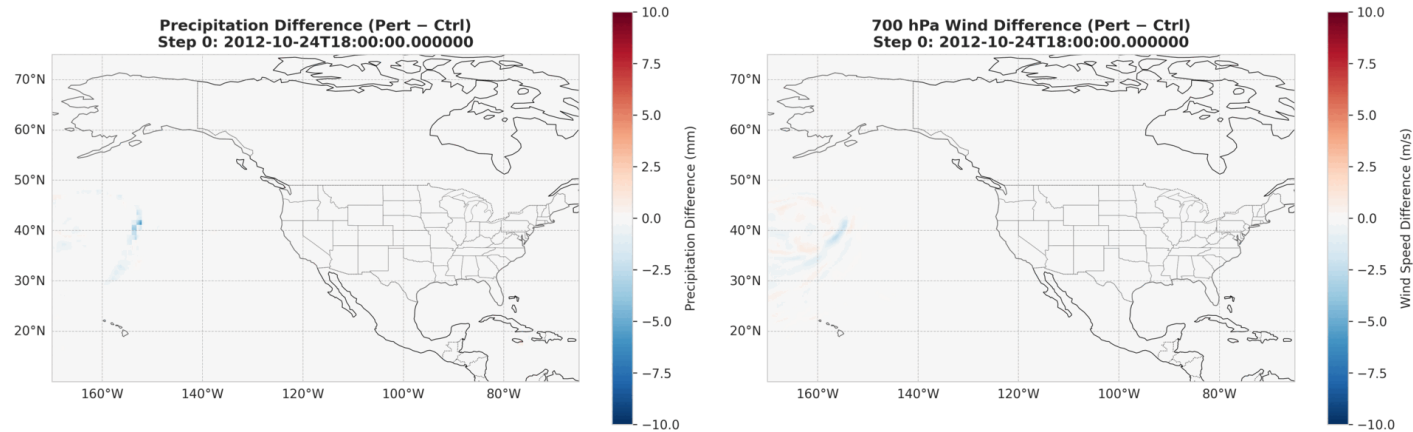
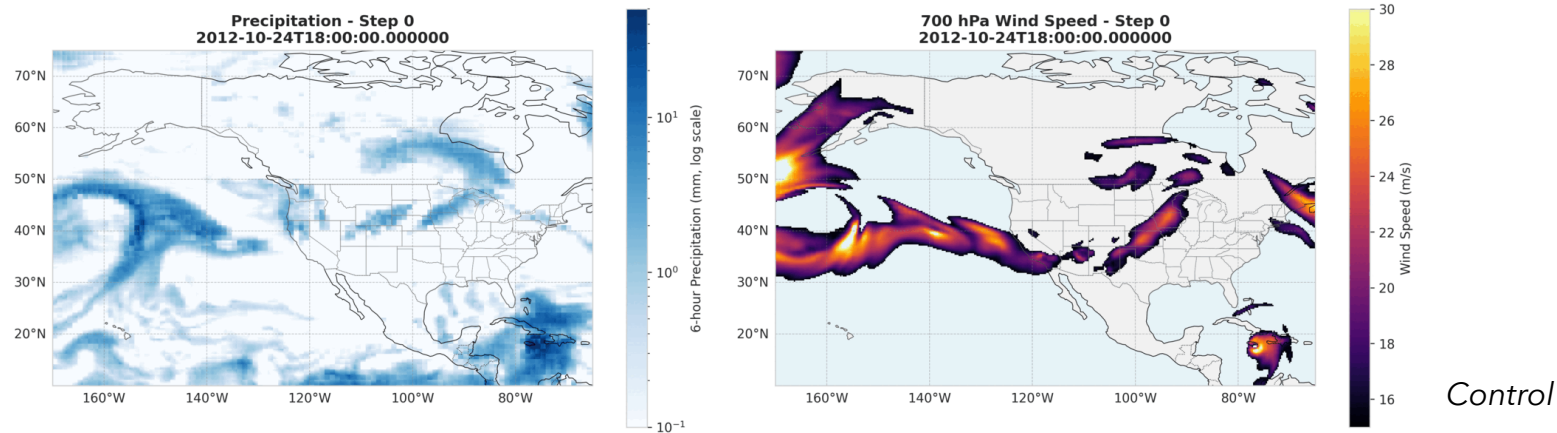
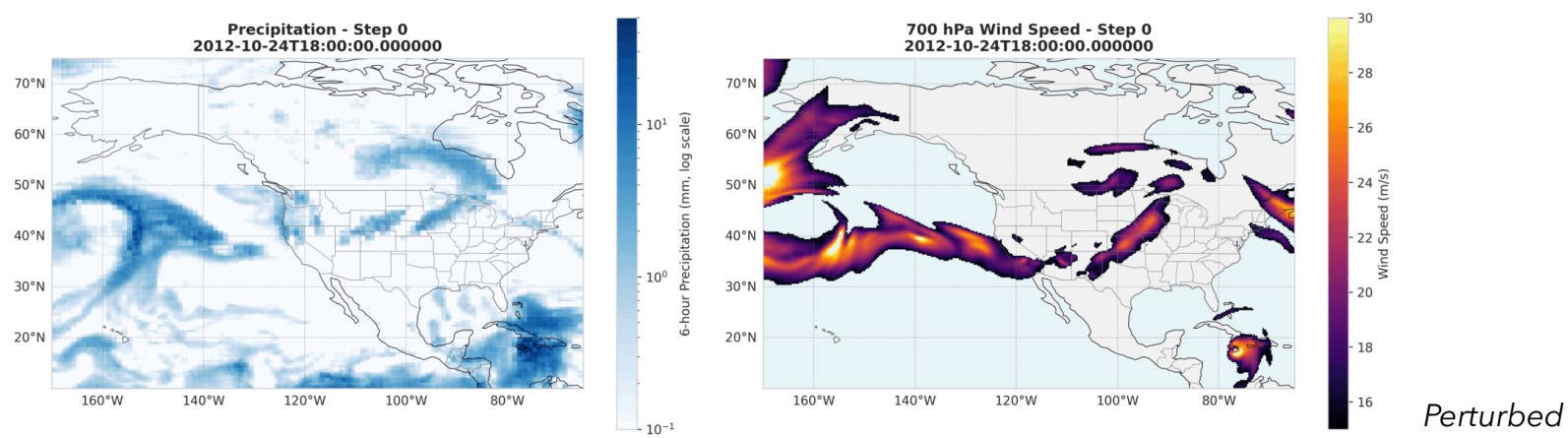


Hurricane Sandy - Perturbation Locations (Pacific Focus) - 2012-10-31
Wind Speed 15-30 m/s (Rainbow, Red=High)

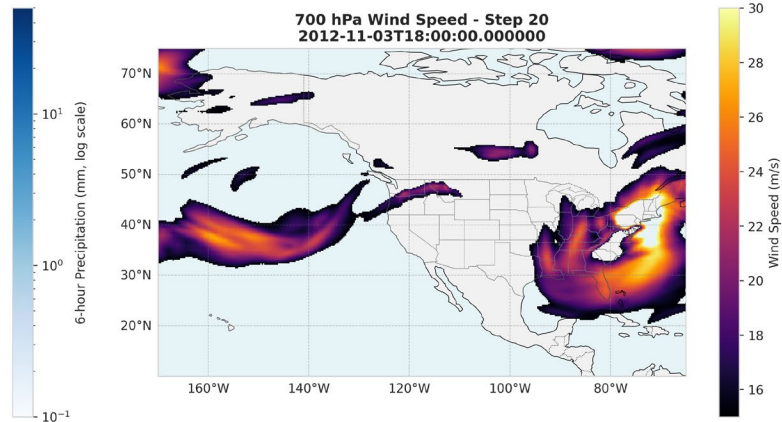
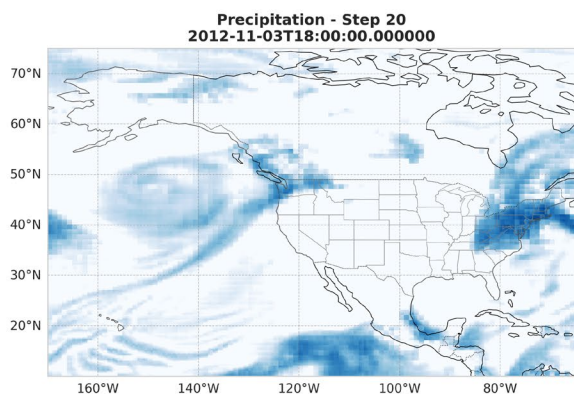


Multiple Perturbations at Single Point

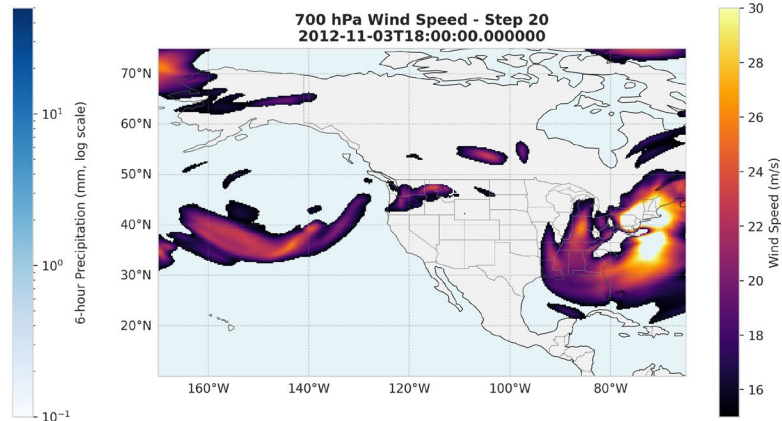
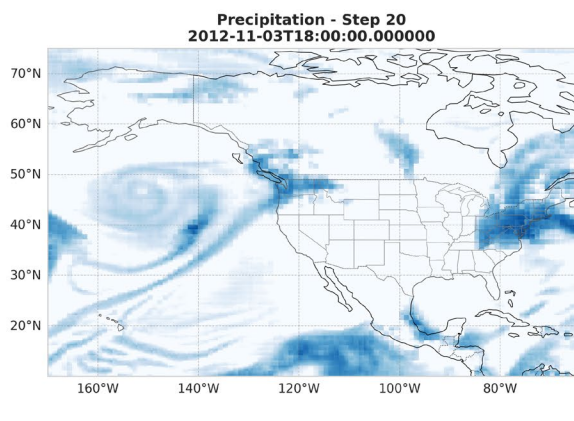




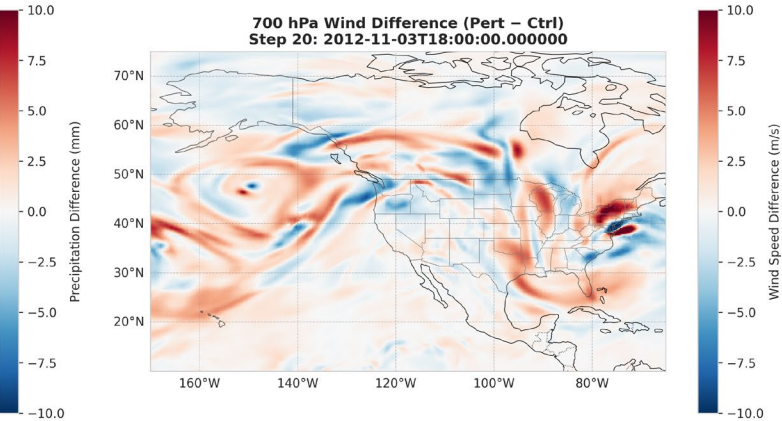
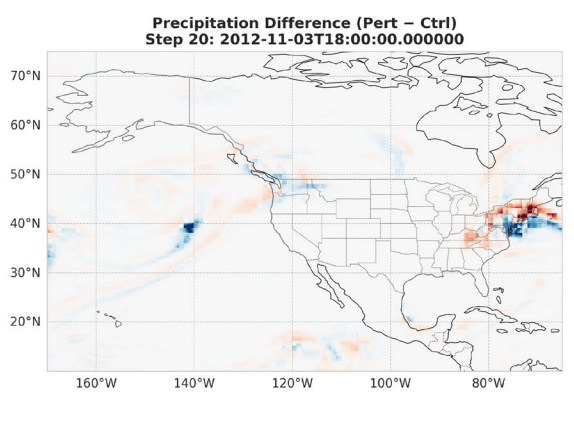
- 170 W 35 N (one of the perturbation points that showed higher effects)
- Perturbation every step



Perturbed



Control



Potential Tasks

- Come up with “better” multi-perturbation schemes
- Try more locations (both single point and multipoint)
- Make meaningful assessments
- Try different storms? (that interacts more with the jet stream)